

Education

Madison, SD	Dakota State University	Fall 2023 – Spring 2027
• Bachelor of Science in Computer Science and Cyber Operations, GPA: 3.96 • Dean's List, Honors Program, DSU Presidential Champion scholarship • Coursework: Data Structures, Advanced Data Structure, Parallel Computing, Networking, Object Oriented Programming, Web Development, Software Security, Assembly Language, Introduction to Cyber Security		

Research Experience

Quantum Security Researcher	Dakota State University	Summer 2024
Multi-level Post Quantum Encryption for Images with Quantum Fourier Transform		
• Designed novel multi-level post quantum encryption algorithm combining QFT, AES, and LWE cryptography, achieving PSNR >40 dB and SSIM 0.95-0.98 while maintaining encryption times under 2.4s for images • Published peer reviewed research in IEEE Xplore		

Technical Experience

Projects

- **Zig Data Structures & Algorithms Library**
 - Implemented comprehensive educational library featuring 10 data structures and 12 algorithms in Zig; designed for systems programmers learning the language
- **Research Article Discussion Platform**
 - Built collaborative web application enabling users to discuss academic papers in real time chat rooms organized by DOI.
 - Implemented user authentication, WebSocket based messaging, and DOI validation system in team of 4 developers
- **Lock Free Concurrent Data Structures**
 - Implemented lock free queue and stack using atomic operations, achieving 4x throughput improvement over mutex based implementations under high contention (8+ threads)
- **Custom Memory Allocator**
 - Built custom memory allocator with segregated free lists targeting small uniform allocations, reducing fragmentation by 35% compared to naive implementations; achieved 2.3x speedup over glibc malloc for game-like workloads while maintaining comparable performance for mixed workloads with 512KB+ allocations.
- **Open Source:** Contributor to Zig programming language

Work Experience

STEM Academic Tutor	Dakota State University	Fall 2024 – Spring 2025
• Mentored GrowStem scholarship recipients in mathematics and programming fundamentals • Guided students through initial phases of undergraduate research project development		

Additional Experience

- **Leadership:** Vice President, English Club; Honors Student Advisory Council
- **Public Speaking:** Multiple Best Delegate awards, Model United Nations competitions

Languages and Technologies

- **Languages:** C (expert), Zig (expert), Python (proficient), SQL (proficient), Java (intermediate), C# (intermediate), C++ (prior experience)
- **Technologies:** Linux, Vim, Git, Spring Boot, SQLite, PostgreSQL